

Client Statistical Analysis Report

Client: Portfolio Demo / Custom CSV Project | **Prepared by:** Tianyou Liu - Statistical Analysis | **Date:** June 18, 2026

Executive Summary

High-level decision summary for business stakeholders before the full statistical methodology.

TEST	Chi-Square Test of Independence
RESULT	No statistically significant difference was detected. $p = 0.3841$ at $\alpha = 0.05$.
RECOMMENDATION	No reliable association was detected; a standardized offer or layout is reasonable.
CONFIDENCE	Degrees of freedom: 2
PLAIN ENGLISH	The Chi-Square Test of Independence did not find a statistically significant result ($p = 0.3841$, $\alpha = 0.05$). The observed difference may be random noise.

1. State

Define the hypotheses and significance level to evaluate the claims of the A/B test.

Significance Level (α): 0.05

Hypotheses (Symbols):

H_0 : Variables are independent H_a : Variables are dependent

Null Hypothesis (H_0): The categorical variables are independent; the distribution of responses is similar across groups.

Alternative Hypothesis (H_a): The categorical variables are dependent; the distribution of responses is significantly different across groups.

2. Plan

Identify the appropriate statistical procedure and verify its mathematical assumptions.

Recommended Procedure: Chi-Square Test of Independence

Assumption / Condition Checked	Status & Diagnostic Details
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Randomness	Categorical frequencies collected from random sampling.
Independence	Observations represent mutually exclusive categorical responses.
Large Sample (Expected counts)	All expected cell counts ≥ 5 ? Yes (Minimum expected cell count = 22.86).

CONDITIONS MET

All mathematical conditions for the Chi-Square Test of Independence are satisfied. Parametric results are highly reliable.

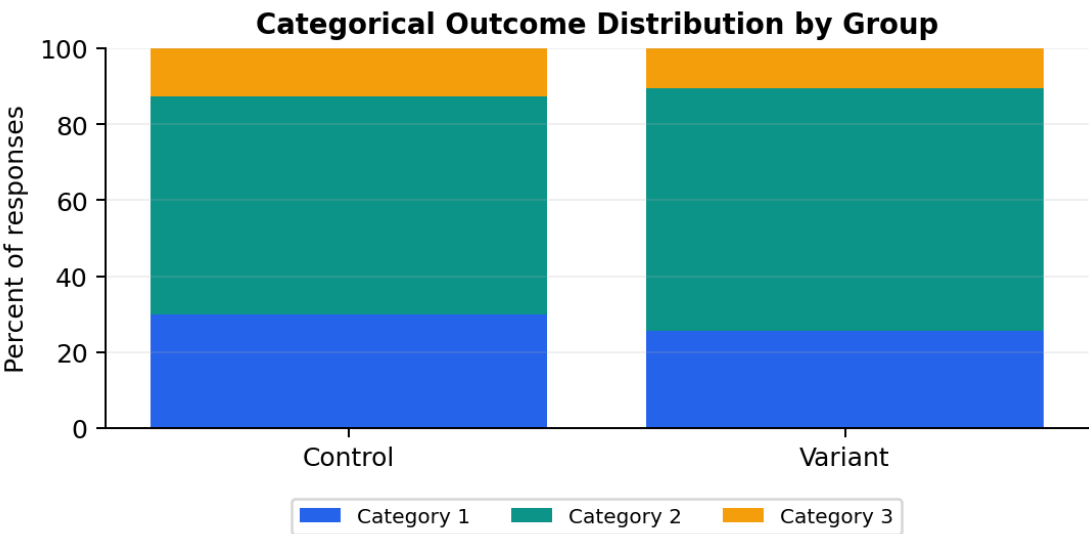
3. Do

Calculate sample metrics, standard error, test statistic, and the resulting p-value.

Sample Descriptive Statistics:

Cell Index (Row, Col)	Observed Count (O)	Expected Count (E)	Residual Contribution (O-E) ²
Row 1, Col 1	60	55.24	0.4105
Row 1, Col 2	115	121.90	0.3911
Row 1, Col 3	25	22.86	0.2009
Row 2, Col 1	56	60.76	0.3732
Row 2, Col 2	141	134.10	0.3555
Row 2, Col 3	23	25.14	0.1826

Visual Summary:



Calculations & Mathematical Formulations:

• *Chi2 Stat:*

$$\chi^2 = \sum \frac{(O - E)^2}{E} = 1.9138$$

• *Df:*

$$df = 2$$

Statistical Parameter	Value
Test Statistic (χ^2)	1.91385
p-value	0.38407
Degrees of Freedom (df)	2
Cramer's V	0.0675 (Association strength for categorical variables)

4. Conclude

State the final decision and offer a business interpretation based on the statistical results.

Compare p-value to α : $0.38407 \geq 0.05$. Since the p-value is greater than or equal to the significance level, we **fail to reject the null hypothesis (H_0)**. There is insufficient evidence to conclude there is a significant difference between the variations.

DECISION: FAIL TO REJECT NULL HYPOTHESIS

Business Action Recommendation: No significant association was detected. User distributions are statistically homogeneous across groups. Standardize the layout/offering.

Plain English:

The Chi-Square Test of Independence did not find a statistically significant result ($p = 0.3841$, $\alpha = 0.05$). The observed difference may be random noise.